Knowledge Verification From Data

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***Abstract*— Knowledge verification is an important task in the quality management of knowledge graphs (KGs). Knowledge is a summary of facts and events based on human cognition and experience. Due to the nature of knowledge, most knowledge quality (KQ) management methods are designed by human experts or the characteristics of existing knowledge, which may be limited by human cognition and the quality of existing knowledge. Numerical data contain a wealth of potential information that may be helpful in verifying knowledge, which is rarely explored. However, due to the implicit representation of numerical data to facts as well as the noise in the data, it is challenging to use data to verify the knowledge. Therefore, this article proposes a knowledge verification method, which discovers the correlation and causality from numerical data to validate knowledge and then evaluate the quality of knowledge. Moreover, to address the impact of noise, the method integrates multisource knowledge to jointly evaluate the KQ. Specifically, an iterative update method is designed to update KQ by utilizing the consistency between multisource knowledge while designing knowledge verification factors based on data causality and correlation to manage update process. The method is validated with multiple datasets, and the results demonstrate that the proposed method could evaluate KQ more accurately and has strong robustness to noise in the data.**

***Index Terms*— Knowledge graph (KG), knowledge management, knowledge quality (KQ), multisources.**

I. Introduction

RECENTLY, the research of knowledge graphs (KGs) has received widespread attention [1], [2]. In this context, knowledge verification, i.e., to confirm whether the description of knowledge to facts is correct, is essential to the utility of KG applications. Generally speaking, knowledge is a summarization of human experience [3], and it is usually acquired from textual data [4] or domain experts [5]. Many existing methods use the knowledge extracted from human experience to evaluate knowledge quality (KQ) [6], [7], [8], [9]. However, the summarization of knowledge by human is time-consuming and may be subjective in some cases, especially in new domains that are not fully explored. It may result in the validation of knowledge being limited to human cognition or correctness of textual data, which makes it difficult to objectively evaluate the KQ.

Compared with knowledge,[1] data are relatively inexpensive to acquire in some domains [10], and it is more objective in presenting facts. It may contain a lot of potential, even undiscovered knowledge, which is not recorded or recognized by humans [11]. If the latent knowledge in data could be mined and used to verify the knowledge in KGs, it may make the evaluation of KQ more objective, which has not been widely explored. However, this process presents challenges.

First, data and knowledge present facts in different forms. Knowledge describing a fact usually consists of entities with specific meanings and the relationships between them in KGs, while in the data, facts are usually presented as a large amount of data. Second, the knowledge contained in data is latent. For example, in many domains, facts in data need to be summarized through the analysis of domain experts. These make it difficult to verify knowledge directly with data. Therefore, to realize the interaction between data and knowledge, it is essential to explore more explicit information from data in the same form as knowledge, that is, "data-driven knowledge."

Furthermore, data may contain a lot of noise [12], which may lead to the extraction of incorrect "data-driven knowledge." This incorrect knowledge may conflict with the knowledge in the KG to be verified. If it is directly used to verify the knowledge, it may result in obtaining the wrong quality of knowledge. Therefore, how to quantify the quality of knowledge in KGs by using the knowledge in data is also a challenge.

Correlation and causality characterize association patterns between entities, which can be extracted from data. Correlation represents the properties are related, while causality [13], [14], [15], [16], [17] represents the causal logical relationship between properties. The correlation and causality between properties are similar to the representation of the relationship between entities in the triple, and some researchers regard it as a kind of knowledge [18], [19], [20]. Generally speaking, when data and knowledge describe the same fact, the knowledge may be considered as high quality (correct) when it agrees with the data. For example, for the triple in a KG: (temperature, influence, and virus activity), if the data show that the virus activity has an obvious causal relationship or correlation with

Manuscript received 2 April 2022; revised 3 August 2022; accepted 22 August 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2021ZD0111700, in part by the National Nature Science Foundation of China under Grant 62137002 and Grant 62176245, in part by the Key Research and Development Program of Anhui Province under Grant 202104a05020011, in part by the Key Science and Technology Special Project of Anhui Province under Grant 202103a07020002, and in part by the Fundamental Research Funds for the Central Universities. *(Corresponding author: Huanhuan Chen.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TNNLS.2022.3202244.

Digital Object Identifier 10.1109/TNNLS.2022.3202244

[1]The knowledge refers to the triples in KGs.

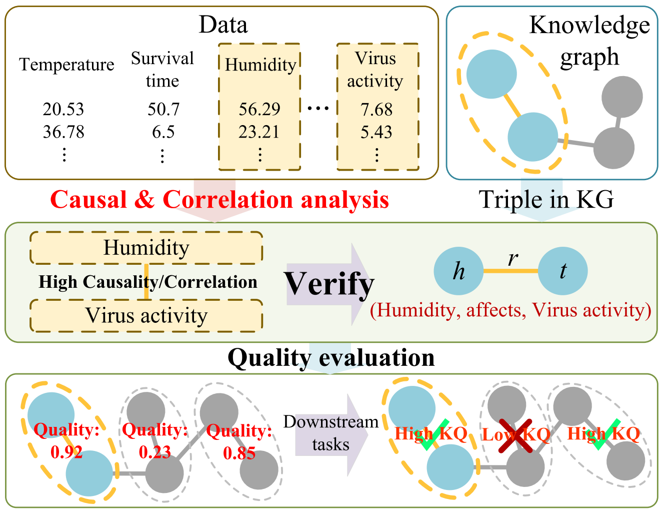

Fig. 1. Knowledge quality evaluation integrating the verification from data. *The upper parts are the utilized data and KG where the data properties such as "temperature" are all included as entities in the KG.

temperature, the triple may be considered as high quality. On this basis, this article integrates the correlation and causality of data to verify the quality of triples, which employs the significance level and causal effect between the properties to evaluate the agreement degree between knowledge and data related to the same fact (as shown in Fig. 1).

In addition, due to the influence of data noise on the performance of existing causal inference algorithms, the obtained causality between properties may not be guaranteed. Therefore, based on knowledge verification from data, this article integrates the consistency of multisource knowledge to evaluate KQ. It evaluates the KQ by using the consistency of the investigated knowledge and knowledge in the external KGs, thereby reducing the impact of data noise. An iterative method is proposed to update the quality of triples in all KGs simultaneously. In each iteration, the quality of knowledge is updated based on the consistency of the investigated knowledge with external knowledge. Then, a causality-verified factor is designed to constrain the updating of KQ, thus reducing the interference of knowledge inconsistent with the data or low-quality external knowledge. The framework of the proposed method is shown in Fig. 2.

The contributions of this article are summarized as follows.

1) This article introduces knowledge verification from data into the KG quality management task. To the best of our knowledge, this is the first work of utilizing data for knowledge verification.
2) A method is proposed to convert the correlation and causality of properties in data into factors for evaluating the quality of triples.
3) The verification of multisource knowledge is integrated with data, which eliminates the interference of noise and incorrect external knowledge in multisource knowledge on KQ evaluation.

The rest of this article is organized as follows. After introducing the background in Section II, the proposed method is illustrated in Section III. Experimental results and discussions are presented in Sections IV and V, respectively. Finally, Section VI concludes this article.

II. Background

A. Motivation: Knowledge Validation From Numerical Data

In recent years, with the widespread applications of KGs, the quality management of KGs has received extensive attention [21]. As an important step of KG quality management, the KQ evaluation could provide a quality quantification and estimation for knowledge and then provide a reliable basis for the improvement and correction of KG. Many quality management methods of KG could be implemented to evaluate the quality of knowledge [22], [23], [24]. For example, Xue and Zou [25] summarized the main types of current knowledge management methods, including human-based [26], statistical and learning of graph pattern-based [27], rule-based [28], and hybrid methods. Most of these methods essentially verify knowledge based on expert knowledge or rules or design methods to manage KQ based on the pattern of the KG itself. Some external "data-assisted" methods are mostly based on textual data or knowledge bases, which, as a form of human recording knowledge, are essentially the same as expert knowledge.

Unlike human experience or existing knowledge, data are defined as a set of discrete, objective facts about events [29], usually in the form of statistics to describe facts in the real world. These statistics exist in large quantities in reality and contain rich latent knowledge, which have great potential to assist knowledge engineering-related tasks. Therefore, aiming at the task of KQ evaluation, this article explores a new method of KG quality management from the perspective of data, different from the existing quality management methods.

B. Correlation and Causality of Data

Correlation represents that the observed results of two events are related to some extent [14], which is commonly assessed through correlation coefficients, statistical independence, and conditional independence (CI) tests [30]. To explore correlation in real-world data, the type of data is essential for analysis approaches, including continuous and discrete data. To deal with different types of data, there are various approaches of independence tests. For example, λ^2 test, G^2 test, and mutual information calculation can be used for discrete data [31]. For continuous data with a linear relationship with Gaussian error, the Fisher Z test could be used [32]. For continuous data with nonlinear and non-Gaussian noise, a kernel-based CI test method could be employed [33].

Causality is explained as cause and effect where the cause is partly responsible for the effect, and the effect is partly dependent on the cause [34]. Causality in the data is widely used in tasks such as feature selection and anomaly detection [35], [36], [37]. To infer the causality, constraint- and score-based methods are mainly applied [38]. Constraint-based methods utilize CI tests to determine whether there is causality between properties. Parents and children (PC) algorithm [39] and fast causal inference (FCI) [40] are typical constraint-based algorithms. Score-based methods select the best matching causal

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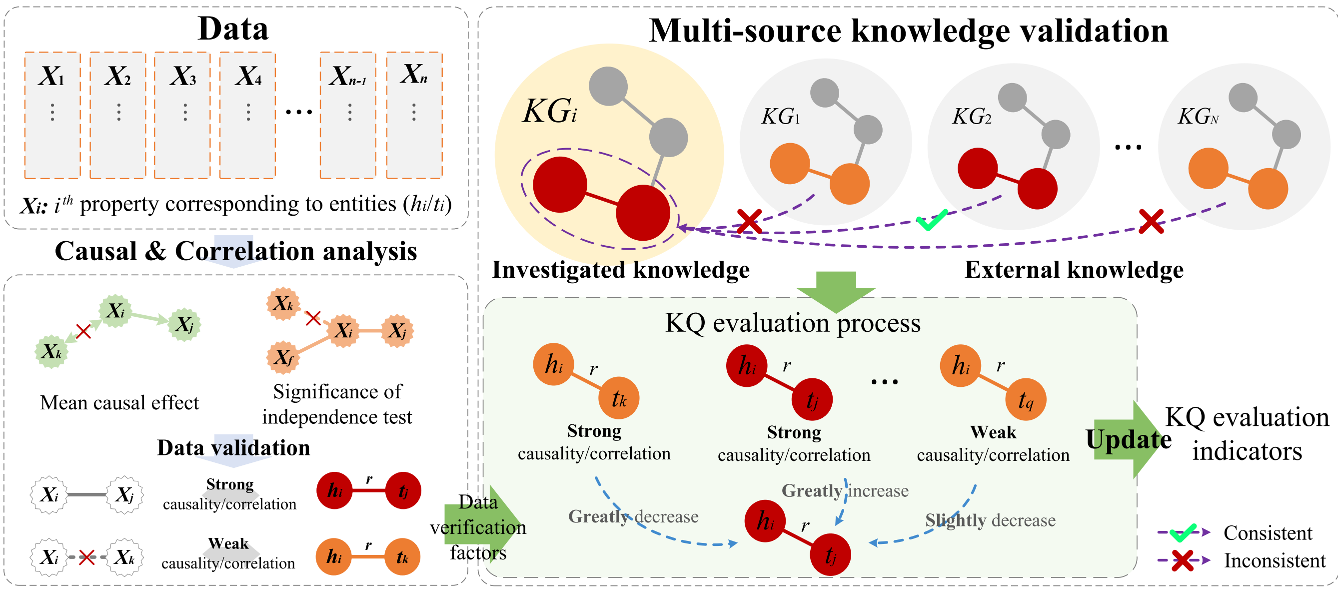

Fig. 2. Knowledge quality evaluation integrating the verification of multisource consistency and causality. *The left part evaluates the causal and correlation degree between entities based on data analysis. The right part evaluates KQ based on the agreement (consistency) of multisource knowledge, where the subframe "KQ evaluation process" integrates the agreement results of multisource knowledge and the causal and correlation degree between entities from data to validate knowledge.

graph from candidate graphs in an optimized way by defining some scoring functions. Greedy equivalence search (GES) [41] is a well-known two-stage process in which the operator used for greedy search yields could be scored efficiently.

In addition, there are other causal inference methods, such as asymmetric-based, sampling-based, and hybrid methods [42]. Asymmetry-based methods exploit the asymmetry according to additional assumptions to determine the direction of the structural relationship [43], [44]. Sampling-based methods confirm the correctness of the graph structure through posterior sampling [45], [46]. Hybrid methods combine ideas from other methods. For example, max-min hill climbing (MMHC) combines constraint- and scoring-based method [47]. This method first learns a causal undirected graph through a local structure learning algorithm and then orients the undirected graph with a greedy Bayesian scoring hill-climbing search method.

The causality could be qualitatively inferred by the above work, while the degree of causality can be obtained quantitatively by calculating the causal effect [13], which requires the causal relationship to be fully or partially known. Due to differences in frameworks and assessment methods, causal effects may be defined and assessed differently. A widely used average causal effect (ACE) is defined as the difference in expected output at different intervention values [48], which is based on the definition of structural causal model. In addition, there are some studies that give definitions and evaluation methods for causal effects, including study of matching-based methods [49], study of tree-based methods [50], and study of ensemble-based methods [51].

C. KQ Evaluation Based on the Consistency of Multisource Knowledge

For the verification of knowledge, a common idea is to compare investigated knowledge with external knowledge to verify the quality of knowledge [52], [53]. Many methods utilize external knowledge bases or external high-quality KGs to validate the knowledge in KGs. This type of method utilizes the consistency of the investigated knowledge with external knowledge to verify knowledge, which usually assumes that a piece of certain knowledge is of high quality when it is consistent with external high-quality knowledge or consistent with a large amount of external knowledge. This consistency could be used to detect incorrect knowledge or further construct quality metric in many tasks. For example, Tao *et al.* [54] evaluated the KQ according to consistency. Considering that the quality of knowledge sources may be different, the author assigns different weights to the knowledge according to the expertise of the knowledge sources. Gerber *et al.* [55] found trustworthy sources on the web to validate knowledge. The authors presented a framework, which gives the input knowledge high quality if many evidences can be found from web pages to support this knowledge. Yin *et al.* [56] proposed an algorithm, which evaluates KQ using multiple knowledge sources. According to the comparison result between knowledge, this algorithm updates the quality of knowledge and knowledge sources iteratively. In addition, the work in [57] and [58] implies similar ideas, which also assesses the quality of knowledge based on the comparison of knowledge in different knowledge sources.

III. Multisource Knowledge Quality Evaluation Based on Data Validation

A. Overall

This section will elaborate the proposed multisource KQ evaluation method based on data validation. For the triple describing a fact, the method utilizes its consistency with external triples related to the same fact from different KGs to verify its quality. An iterative update process is

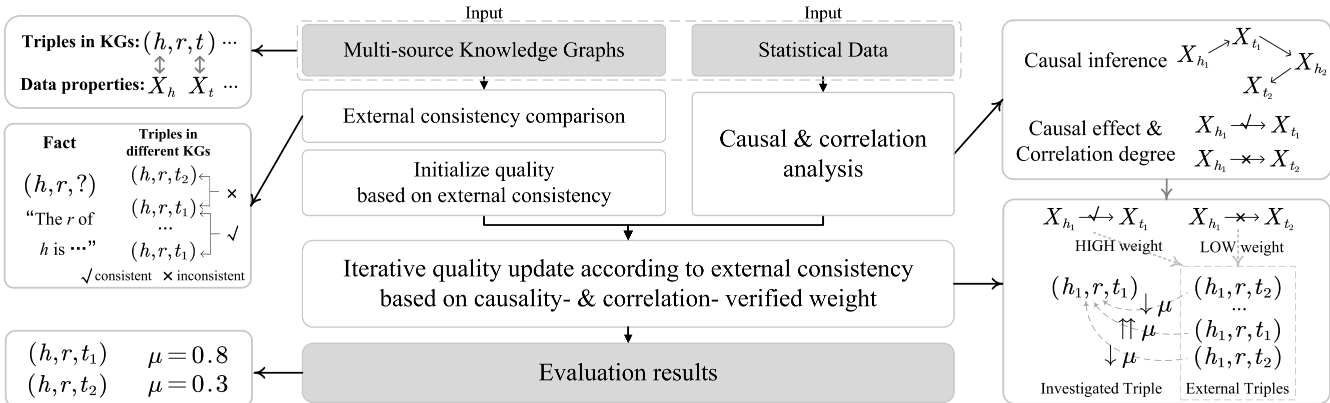

Fig. 3. Flowchart of the proposed method.

proposed to evaluate the quality of each triple in each KG. This process uses external triples to confirm the quality of investigated triple, where the quality of the external triple and its agreement degree with data are regarded as the weight. For data validation, correlation- and causality-verified factors are constructed using the correlation and causality of data corresponding to triples as part of the weight. The flowchart of the proposed method is shown in Fig. 3. Specifically, necessary notations, assumptions, and definitions are presented in Section III-B. Second, triple quality updating based on multisource knowledge is introduced in Section III-C. Next, Sections III-D and III-E introduce data correlation and causality for knowledge validation, respectively.

B. Preliminaries

1) Notations, Assumption, and Problem Setting: Notation 1 (KGs and Triples): Suppose that there are N KGs totally, expressed as $\mathcal{K} = \{\mathcal{K}_i | i \in \{1,2,\ldots,N\}\}$. Each KG $\mathcal{K}_i$ contains a set of triples, denoted as $\mathcal{K}_i = \{k_i^1, k_i^2, \ldots, k_i^n\}$. k_i^f represents the triple describing fact f in KG $\mathcal{K}_i$, which has the form of (h_i^f, r_i^f, t_i^f). Assume that E is the set consisting of all entities that occur in the KG.

Notation 2 (External Triples): Given the investigated triples k_i^f, k_e^f is utilized to represent the external triple of k_i^f that describes the same fact[2] f in KG $\mathcal{K}_e$ (where $e \in \{1,2,\ldots,N\}, e \neq i$). All the external triples of the investigated triple k_i^f could be grouped and expressed as $E_i^f = \{k_e^f | e \neq i\}$.

Remark 1: Since the proposed method is equivalent for each fact f, the following discussion may omit the superscript f for concise illustration. In particular, k_i^f and k_e^f could be expressed as k_i and k_e, respectively, and (h_i^f, r_i^f, t_i^f) could be expressed as (h_i, r_i, t_i) in the following.

Notation 3 (Knowledge Quality): The KQ reflects the degree to which knowledge conforms to facts [21], [59]. An indicator is constructed to quantify the KQ, denoted as $\mu \in (0,1)$.

[2]In this article, triples with the same head entity and relation are regarded to describe the same fact. For example, the triples (Newton, be-born-in, 1643), (Newton, be-born-in, 1644), and (Newton, be-born-in, 1645) describe the same fact.

Notation 4 (Data Matrix): Data in this article are expressed as a matrix $D = (X_1, X_2, \ldots, X_\alpha, \ldots, X_m)$, where each X_α represents a property and there are a total of m properties. In addition, this article defines pv_h as the total number of different possible values of X_h, and supposes that $\{V_h(1), \ldots, V_h(\mathrm{pv}_h) | \forall i \neq j, V_h(i) \neq V_h(j)\}$ are different possible values of property X_h.

Assumption 1: Given that h (or t) is an entity in KG, this article assumes the property that records the value of h (or t) that occurs in the data matrix, represented as X_h (or X_t).

Problem Setting: Given the set of KGs $\mathcal{K}$ and data matrix D, the goal of this article is to evaluate the quality μ_i^f of triple k_i^f related to each fact f in each KG $\mathcal{K}_i (i = 1,2,\ldots,N)$.

2) Correlation and Causality: A causality-based KQ evaluation method is proposed in this article. To introduce the proposed method, some definitions about correlation are necessary.

Definition 1 (G^2 Test): G^2 test is used as a method of CI test to detect the correlation between two entities. Given X_h and X_t, the G^2 statistic is defined as

$$G^2 = 2\sum_{a=1}^{\mathrm{pv}_h}\sum_{b=1}^{\mathrm{pv}_t} O(a,b)\ln\left(\frac{O(a,b)}{H(a,b)}\right) \tag{1}$$

where $O(a,b)$ is the number of samples in which the property X_h valued as $V_h(a)$ and the property X_t valued as $V_t(b)$. X_h valued as $V_h(a)$ and X_t valued as $V_t(b)$ when h and t are independent.

The significance level could be obtained according to the G^2 statistic, which is the probability of rejecting the null hypothesis given that the null hypothesis is true [60]. In the case of CI test, it reflects the degree to which entities are independent of each other.

Definition 2 (Significance Level [60]): Suppose that $\alpha(X_h, X_t)$ represents the significance level for X_h and X_t, which could be got via checking the cutoff value table of the chi-square distribution with specific degrees of freedom [61]. For X_h with pv_h possible values and X_t with pv_t possible values, the degrees of freedom are $(\mathrm{pv}_h - 1) \times (\mathrm{pv}_t - 1)$.

Definition 3 (ACE [48]): ACE is the difference in expected output at different interventions. Specifically, the expected difference on the probability of X_t valued as $V_t(b)$, when

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intervenes property X_h valued as $V_h(a_2)$ by $V_h(a_1)$, could be expressed as

$$\begin{aligned} &ACE(V_t(b), V_h(a_1), V_h(a_2)) \\ &\quad = P(x_t = V_t(b)|\mathrm{do}(x_h = V_h(a_1))) \\ &\qquad - P(x_t = V_t(b)|\mathrm{do}(x_h = V_h(a_2))) \end{aligned} \tag{2}$$

where x_h and x_t represent the corresponding random variables of properties X_h and X_t, respectively. Operation do$(\cdot)$ indicates an intervention. For example, do$(\cdot)$ in $P(x_t = V_t(b)|\mathrm{do}(x_h = V_h(a)))$ represents to calculate the probability value of $x_t = V_t(b)$ under conditions of $x_h = V_h(a)$ and delete the edge pointing to h in the causal graph (thus removing the corresponding causal relationship).

C. KQ Evaluation Incorporating Multisource Consistency and Verification From Data

For multisource KQ evaluation, a classic method is to use external consistency to evaluate the quality of investigated knowledge. This approach assumes that the triple could be considered high quality if it agrees with many external triples in other KGs. The consistency of multisource knowledge is employed to evaluate KQ. An iterative update method is designed, which updates the KQ through external consistency comparison and correlation and causality. The proposed method is presented as follows:

$$\mu_i^{l+1} = \mathcal{I}\left(\mu_i^l; \left\{\mu_e^l, R_{k_e}, S_{k_e}, c_e | k_e \in E_i\right\}\right) \tag{3}$$

where superscript l represents the lth iteration, μ_i^{l+1} is the quality of the investigated triple k_i, and μ_e^{l+1} is the quality of external triple k_e obtained in the lth iteration. $\mathcal{I}(\cdot)$ denotes the proposed updating method in each iteration. R_{k_e} and S_{k_e} represent the correlation and causal strength of triples k_e by analysis from data, respectively.[3] Note that only the calculation of μ_i^{l+1} is introduced and μ_e^{l+1} is calculated exactly same as μ_i^{l+1}. Each triple is taken as the investigated triple k_i and its quality in the next iteration is calculated based on the quality of all triples in the previous iteration. μ_e^l and μ_i^l are only symbols to distinguish external triple and investigated triple where e and i are both indexes with no fundamental difference.

c_e is utilized to represent the consistency comparison results, obtained by comparing the investigated triple k_i with its external triple k_e. c_e is defined as

$$c_e = \begin{cases} 1 & k_e = k_i \\ -1 & k_e \neq k_i \end{cases}, \quad k_e \in E_i \tag{4}$$

where $k_e = k_i$ ($k_e \neq k_i$) means that triple k_i is consistent (inconsistent) with external triple k_e. For the sake of illustration, this article judges the consistency by checking whether the tail entities of two triples are the same during the comparison process. For example, investigated triples (h, r, t_i) and (h, r, t_e) are considered consistent if $t_i = t_e$; otherwise, they are inconsistent.

Each triple k_i describing the same fact is updated synchronously in each iteration. For triple k_i, the update of its quality μ_i is expressed as

$$\mu_i^{l+1} = \mu_i^l + B^l \times P^l \tag{5}$$

[3]The definitions and utilization of R_{k_e} and S_{k_e} are introduced in Sections III-D and III-E, respectively.

where $B^l \in (0, 1)$ represents the upper bound and $P^l \in (-1, 1)$ represents the update proportion, which integrates consistency comparison, correlation, and causality. The direction of update is determined by P^l, and μ_i^{l+1} increases if and only if P^l is positive.

The definition of update proportion P^l is

$$P^l = \frac{\sum_e c_e W_e^l}{|E_i|} = \frac{\sum_{e,t_e=t_i} W_e^l - \sum_{e,t_e\neq t_i} W_e^l}{|E_i|} \tag{6}$$

where $|E_i|$ is the number of external triples. W_e^l is the weight of triple k_e in the lth iteration, representing how much it influences the quality updates of investigated triples in the consistency comparison. The weight is determined by the estimated quality of the triple and data-verified factors, increasing or decreasing the quality of the investigated triple based on the consistency result c_e. When the external triple is consistent with the investigated triple, the investigated triple is affirmed, and the update proportion P^l has a positive change. Also, when they are inconsistent, P^l is relatively reduced.

The definition of W_e^l is presented as follows:

$$W_e^l = g_1\left(\mu_e^l, \mu_i^l, R_{k_e}, R_{k_i}\right) \times g_2\left(\mu_e^l, S_{k_e}\right) \tag{7}$$

where $g_1(\mu_e^l, \mu_i^l, R_{k_e})$ and $g_2(\mu_e^l, S_{k_e})$ are the correlation- and causality-verified factors, respectively, which adjust the weights of knowledge in the consistency comparison by data.[4]

The upper bound B^l is defined as

$$B^l = \begin{cases} \frac{1}{l}\left(1 - \mu_i^l\right) & P^l \geq 0 \\ \frac{1}{l}\mu_i^l & P^l < 0. \end{cases} \tag{8}$$

B^l limits the estimated quality μ_i^l in the range of 0 to 1. Factor $(1/l)$ is a convergence term. It is used to adjust the influence of the comparison results in the KQ evaluation. It is true that the quality of each knowledge gradually approaches its real KQ with the iterative process since the comparison results could reflect the KQ. However, excessive reference to the comparison results will introduce bias, as the comparison results are nonquantitative and can only indicate an update direction in terms of describing KQ. The logic diagram of the proposed iterative KQ evaluation method is shown in Fig. 4.

For the initialization of u_i, it is common that the quality of knowledge is initialized with the overall quality of its source [56], [62], that is, the initial quality for each triple in the KG $\mathcal{K}_i$ could be initialized as the quality of $\mathcal{K}_i$. However, the quality of each KG in multisource scenarios may not be inherently known. Therefore, the quality of a KG is approximated by the correct probability of its containing knowledge that could be estimated by consistency statistics. It could be calculated through the following equation:

$$\mu_i^1 \mu_j^1 + \left(1 - \mu_i^1\right)\left(1 - \mu_j^1\right) = \frac{\left\|\left\{f : k_i^f = k_j^f\right\}\right\|}{n}, \quad i \neq j \tag{9}$$

where μ_i^1 and μ_j^1 are the initial quality of each triple in $\mathcal{K}_i$ and $\mathcal{K}_j$, which represent estimates of the probability that triple is correct. Therefore, the left-hand side of (9) is the probability that the knowledge related to the same fact in $\mathcal{K}_i$

[4]Discussed in Sections III-D and III-E.

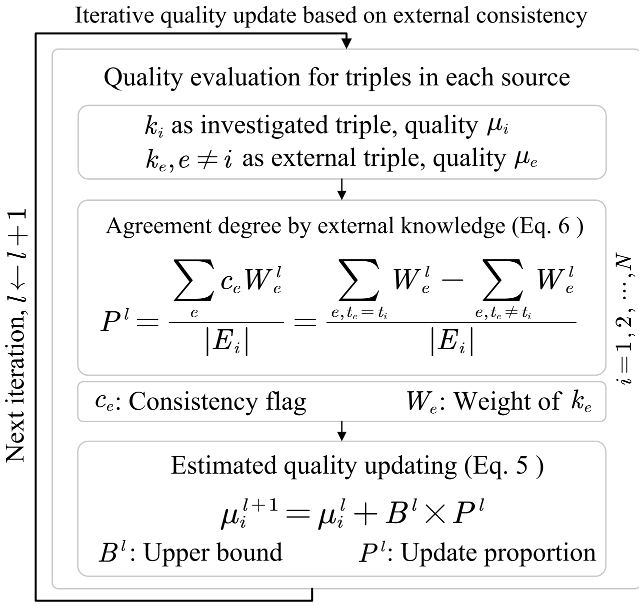

Fig. 4. Logic diagram of the iterative KQ evaluation method based on external consistency.

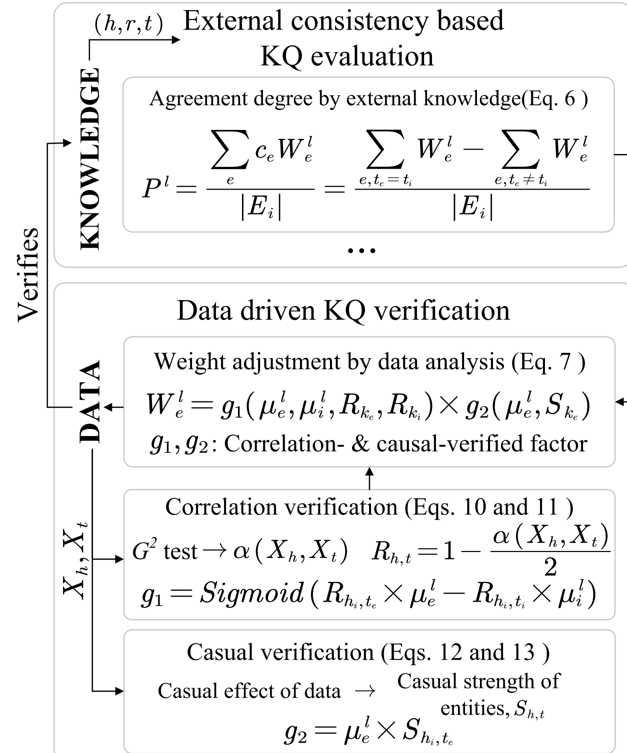

Fig. 5. Logic diagram of the data verified KQ evaluation method.

and $\mathcal{K}_j$ is both correct or incorrect, where the two triples are consistent.[5] $\|\{f : k_i^f = k_j^f\}\|$ is the number of consistent knowledge between $\mathcal{K}_i$ and $\mathcal{K}_j$. The right-hand side represents the probability that knowledge in $\mathcal{K}_i$ and $\mathcal{K}_j$ is consistent, estimated by the ratio of consistent knowledge. To calculate μ_i^0 in multisource scenarios, simultaneous equations could be constructed by combining each two KGs. The initial quality of triples in each KG could be calculated by solving these equations.[6]

D. Correlation-Verified Factor Calculation via Independent Test

The correlation between properties in data could reflect the degree to which their corresponding entities tend to form relationships. Therefore, this article designs a correlation-verified factor to adjust the weight of a triple according to the correlation degree between its entities. The logic diagram of data correlation and causality-verified KQ evaluation is shown in Fig. 5.

The correlation-verified factor $g_1(\mu_e^l, \mu_i^l, R_{k_e}, R_{k_i})$ in (7) is defined as

$$g_1\left(\mu_e^l, \mu_i^l, R_{k_e}, R_{k_i}\right) = \text{Sigmoid}\left(R_{h_i,t_e} \times \mu_e^l - R_{h_i,t_i} \times \mu_i^l\right) \tag{10}$$

where R_{h_i,t_e} is another representation of R_{k_e}, focusing more on the entities. R_{h_i,t_e} represents the correlation between data property X_h and X_{t_e}, corresponding to h and t_e, which are head and tail entities of the external triple. Similarly, R_{h_i,t_i} represents the correlation of head and tail entities of the investigated triple in the corresponding data.

Sigmoid is employed to calculate the relative weight of the external triple based on the difference of external and investigated triples' correlation in data. When the quality of external triple exceeds the investigated triple to some extent, the influence of external triple is activated in this comparison, and the upper bound is larger. When the quality difference is not large, the updated weight is relatively small. Specifically, if they are all of low quality, the comparison weights are not sufficiently instructive. If the investigated triple is high quality and the external is low quality, the influence of external triple is more suppressed.

In the case of inconsistent knowledge during consistency comparison, the entities with higher correlation degree could be considered to form relation more possibly and, thus, more credible from the perspective of correlation analysis. Therefore, the correlation R_{h_i,t_e} is utilized to correct the estimated quality of external triples in (10). The investigated triples are similar.

For the correlation analysis of properties corresponding to the entities h and t of a triple, this article uses the significance level $\alpha(X_h, X_t) \in (0, 1)$ to get the correlation $R_{h,t}$

$$R_{h,t} = 1 - \frac{\alpha(X_h, X_t)}{2}. \tag{11}$$

It is reasonable since the probability that X_h and X_t are related has a negative correlation with the value of $\alpha(X_h, X_t)$, that is to say, when the correlation between X_h and X_t is significant,

[5]Two pieces of knowledge are consistent when both correct, but they may be inconsistent when both false since false knowledge could be diverse. Nevertheless, the desired initial quality could still be approximated through this equation.

[6]In practice, the initial quality of any three KGs could be calculated by constructing three equations between them. The initial quality of the rest KGs could be calculated through combination with any KG in those three.

$\alpha(X_h, X_t)$ is small and the weight of the corresponding knowledge is close to 1. When there is no obvious relationship between the values of h and t, $\alpha(X_h, X_t)$ is large and the weight of the corresponding knowledge in the comparison is only half of the former.

E. Causality-Verified Factor Calculation via Causal Inference

Causality is to describe how causes influence effects, which is a directed relationship that is more specific than correlation. For example, the age of man may be correlated with the disease rate, but it cannot be asserted that older age is the cause of the disease. Moreover, causality is capable to indicate the possibility of relation formation with stronger confidence than correlation. Two events with high correlation could have no relation in reality, whereas events with causality are certainly related to some extent, at least causally related. Therefore, causality is utilized as the main data-verified factor, having more influences on the weight of triples than correlation.

This section introduces the update process that incorporates causality. By transforming the causality between data into a factor that validates knowledge, it realizes the consideration of the causality of investigated triple-related entities in the consistency comparison. The causality-verified factor $g_2(\mu_e^l, S_{k_e})$ in (7) is defined as

$$g_2\left(\mu_e^l, S_{k_e}\right) = \mu_e^l \times S_{h_i,t_e} \tag{12}$$

where S_{h_i,t_e} is another representation of S_{k_e}, representing causal strength between head entity h_i and tail entity t_e corresponding to properties X_{h_i} and X_{t_e} in data.

Similar to correlation, causality could also indicate the possibility of entities to form relations. In comparison to the correlation-verified factor in (10), S_{h_i,t_e} is utilized to correct the quality of the external knowledge and the corrected quality is directly used as the causality-verified factor. Therefore, $(\partial/\partial S_{h_i,t_e})g_2$ is much higher than $(\partial/\partial R_{h_i,t_e})g_1$, indicating that the causality is treated as the main data-verified factor.

In this article, causal strength of a triple is defined as the probabilistic impact of the value change of the head entity on the tail entity, which is an evaluation metric describing the overall causality of triple. This article uses the ACE of the head entity on the tail entity in the knowledge to calculate causal strength.[7]

More specifically, the causal strength $S_{h,t}$ of triple (h, r, t) is expressed as follows:

$$S_{h,t} = \frac{1}{\mathrm{pv}_t} \sum_b \max_{a_1,a_2}(\mathrm{ACE}(V_t(b), V_h(a_1), V_h(a_2))) \tag{13}$$

where pv_t represents the total number of different possible values of X_t and $S_{h,t}$ represents the average effect of the head entity on the value probability of the tail entity.

$S_{h,t}$ has the characteristics of ACE, which are large when the causality of the head entity on tail entity is strong. Specifically, the causal strength $S_{h,t}$ is related to the causal relationship structure between entities. In a causal relationship structure,

[7]The calculation of ACE is based on the causal graph. For the process of discovering causal graph, this article uses MMHC [47], which has been introduced in Section II-B.

Algorithm 1 Knowledge Quality Evaluation via Multi-source Consistency and Verification From Data

Input: knowledge describing the same fact $k_i = (h_i, r_i, t_i)$, $i = 1, 2, \ldots, N$;
Set of entities E;
Number of total iterations L.
Output: $q_1, q_2, \ldots, q_N$, i.e. the knowledge quality of knowledge $k_1, k_2, \ldots, k_N$.

for $h \in E$ **do**
  for $t \in E$ **do**
    `// Correlation degree and Causal strength`
    Calculate $R_{h,t}$ and $S_{h,t}$ by Eqs. (11) and (13) ;
  end
end
Initialize quality $\mu_1^1, \mu_2^1, \ldots, \mu_N^1$ by Eq. (9) ;
for *l=1:L* **do**
  `// In the` l^{th} `iteration`
  for *i=1:N* **do**
    `// Quality update of` k_i
    for $\forall e \neq i$ **do**
      `// Validate` k_i `with` k_e
      $g_1 = Sigmoid(R_{h_e,t_e} \times \mu_e^l - R_{h_i,t_i} \times \mu_i^l)$
      $g_2 = \mu_e^l \times S_{h_e,h_e}$
      $W_e^l = g_1 \times g_2$
      if $k_e = k_i$ **then**
        $c_e = 1$
      else
        $c_e = -1$
      end
    end
    Calculate P^l using Eq. (6)
    if $P^l \geq 0$ **then**
      $B^l = \frac{1}{l}(1 - \mu_i^l)$
    else
      $B^l = \frac{1}{l}\mu_i^l$
    end
    $\mu_i^{l+1} = \mu_i^l + B^l \times P^l$
  end
end
for *i=1:N* **do**
  $q_i = \mu_i^{L+1}$;
end
return $q_1, q_2, \ldots, q_N$;

a directed path from a source entity to a target entity illustrates the causal relationship between the source and the target. If there is only one path from the source to the target, the closer the vertices on the path to the target, the stronger the causal relationship to the target. For example, there is a directed path "$e_1 \longrightarrow e_2 \longrightarrow e_3$"[8] and no other path from "e_1" to "e_2." Logically, the statement "e_2 causes e_3" is a more important and convincing explanation of the reason for "e_3" than "e_1 causes e_3." Therefore, the causal strength between

[8]e_1, e_2, and e_3 represent different entities.

"e_1" and "e_3" is lower than "e_2" and "e_3," which satisfies that the causal strength between entities can explain the semantic credibility of the corresponding knowledge to a certain extent.

Furthermore, if there exists more than one directed path from the source entity to the target entity in the causal relationship structure, the situation becomes more complicated. For example, there are two directed paths "$e_1 \longrightarrow e_2 \longrightarrow e_3$" and "$e_1 \longrightarrow e_4 \longrightarrow e_3$" in the causal relationship structure. Logically, "e_1" indirectly influences "e_3" in two different ways, while "e_2" and "e_4" are direct but partial causes of "e_3," and it is difficult to say which statement is the best of "e_1 causes e_3," "e_2 causes e_3," and "e_4 causes e_3." Therefore, the credibility of the three statements depends on specific situations. In this case, the order of causal strength between "e_1," "e_2," and "e_4" on "e_3" is also uncertain to be consistent with the aspect of semantic analysis.

At the end of this section, the final formula of P^l is written explicitly to clarify the framework of the proposed method more intuitively. Combined with (6), (7), (10), and (12), the update proportion P^l could be expressed as

$$P^l = \frac{\sum_e c_e \mu_e^l S_{h_i,t_e} \mathrm{Sigmoid}\left(R_{h_i,t_e} \times \mu_e^l - R_{h_i,t_i} \times \mu_i^l\right)}{|E_i|} \tag{14}$$

where c_e is the result of comparison, which comes from multisource knowledge in $\mathcal{K}$, R_{h_i,t_e} and S_{h_i,t_e} are correlation and causality strength, extracted from the data matrix D, respectively, and μ_i^l and μ_e^l approach the KQ in the iterative process, which is initialized by the consistency between KGs. The overall procedure of the proposed method is presented in Algorithm 1.

IV. Experimental Results

In this section, experiments are conducted on different datasets to show the effect of data verification on the KQ evaluation. Section IV-A introduces the datasets, including a real-world dataset derived from clinical data of patients with diabetes mellitus type 2 (TD2M), and a simulated dataset generated based on a Bayesian network in [63]. Section IV-B illustrates the detailed settings for experiments. In Section IV-C, experiments are conducted from different perspectives to show the effects of different factors on the model performance.

A. Datasets

1) Dataset (A): Binder *et al.* [63] provided a Bayesian network for estimating the expected claim costs for a car insurance policyholder. It is mainly about the status of the car and owner and accident occurrence. For example, the antilock brake system (ABS) of the car could affect the probability of accident occurrence, and their conditional probability is included in the Bayesian network. To simulate the real-world investigation on car owners, the related data are generated based on the conditional probabilities in the Bayesian network, which contains 5000 samples.

As for the KG, the nodes of the Bayesian network are taken as entities, where relations between these entities are manually constructed to generate a KG. This KG, containing 52 triples and 27 entities, is taken as ground truth and multiple KGs are generated based on it, introduced in detail in Section IV-B.

2) Dataset (B): Clinical data of patients with TD2M collected in the Department of the First Affiliated Hospital of the University of Science and Technology of China are utilized as the data that drive the KQ verification [64]. The T2DM dataset contains 776 samples and 43 properties on the basic information and various medical indicators of patients, such as age, gender, and body mass index (BMI).

The related KG is constructed by medical experts on the relation of these data properties, based on which utilized multisource KGs are generated according to the approach in Section IV-B. Note that some patient samples contain several missing values and abnormal values, and the data are cleaned and completed by preprocessing for utilization.

3) Dataset (C): ALARM is a Bayesian network for medical diagnosis, proposed by Beinlich *et al.* [65] and used as alarm message system. Based on the state of some medical symptoms of patient such as left ventricular failure or cardiac output, it calculates probabilities for a differential diagnosis. According to the conditional probability provided by ALARM, the data containing 5000 samples are generated. For the generation of KG, nodes and edges in the Bayesian network are considered entities and relations, respectively, leading to a KG with 37 entities and 46 relations.

4) Dataset (D): Hailfinder, introduced in [66], is a Bayesian system to forecast severe weather in Northeastern Colorado. The entities in Hailfinder are mainly related to meteorology, such as moisture, vertical motion, and other atmospheric conditions. Similar to Dataset (A), 5000 samples are generated as the simulated data and a corresponding KG containing 66 triples and 56 entities is constructed. This KG is utilized as ground truth for experiment on Dataset (D) and multiple KGs are generated based on it.

B. Experimental Setup

To conduct experiments on quality evaluation of triples in multisource KGs, different KGs are generated based on original triples of datasets (A) and (B). A generation approach mentioned in [67] is utilized in this article. To generate a KG, the original triples are regarded as correct knowledge, and each tail entity of an original triple is randomly reset to a different one with a uniform probability p, thus producing incorrect triples in the generated KG. In the experiments, the parameter p is adjusted for each generated KG to simulate different quality conditions of KGs in real-world scenarios. To assess method performances, the triple with the highest estimated quality of triples related to the same fact is predicted as the correct one. The ratio of true positive predictions for the ground truth of facts is utilized as its accuracy.

C. Results and Analysis

In order to demonstrate the effectiveness of the proposed method, the influence of the main factors on the experiment is analyzed from different perspectives, including the representative KQ evaluation results of the proposed method, the performance of the method under different accuracy conditions of data analysis, the impact of noise in data, and the impact of the number of KGs in the multisource scenario.

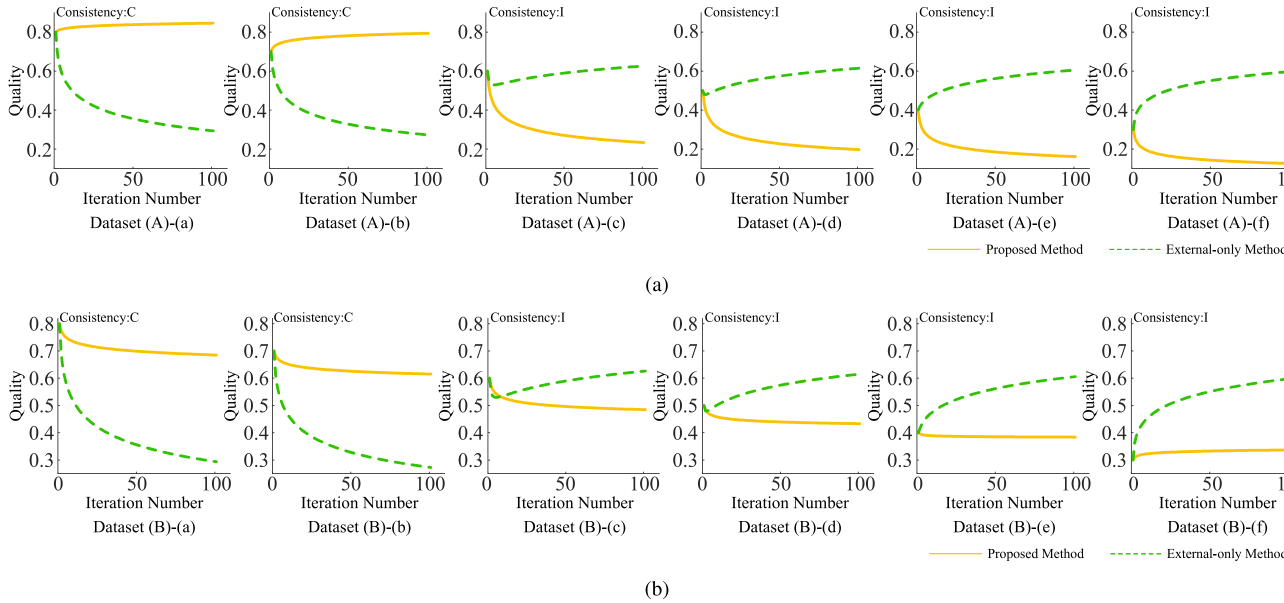

Fig. 6. Comparison between the proposed method and the external-only method. *The "consistency" in each subfigure refers to whether the triple is consistent with the ground truth, where C (or I) represents yes (or no). Note that triples with the same "consistency" state are consistent with each other. (a) Triple quality evaluation in Dataset (A). (b) Triple quality evaluation in Dataset (B).

There are three variants of the proposed method utilized for ablation experiments. They are the external-only method, causal method, and correlation method.

1) For external-only method, the verification from data is deemphasized by setting the correlation and causal strength for all triples to a uniform value 1 ($\forall i \in \{1, 2, \ldots, N\}$, $S_{k_i} = 1$, $R_{k_i} = 1$).
2) The causal method utilizes the causal strength between the head and the tail entity as the quality of a triple, i.e., taking $S_{h,t}$ from (13) as the quality of triple (h, r, t).
3) Similarly, the correlation method utilizes the correlation degree between the head and the tail entity as the quality of a triple, i.e., $R_{h,t}$ from (11).

The external-only method ignores the verification from data and only uses the multisource consistency to evaluate KQ, while causal and correlation methods only depend on the data analysis results but ignore the multisource consistency.

1) Representative KQ Evaluation Results of the Proposed Method: In order to show the effectiveness of the data-verified KQ evaluation, representative results are reported as the form of change trend with iterations. The proposed and external-only methods are compared in evaluating the quality of triples describing the same fact in multisource KGs. Representative results on Datasets (A) and (B) are shown in Fig. 6(a) and (b), respectively. Each subgraph represents the change of the quality of a triple in a KG, where all triples in a group of experiments describe the same fact. For the proposed method, the obtained causal strength of the correct knowledge is 0.96 in Fig. 6(a) and 0.38 in Fig. 6(b), and that of the incorrect knowledge is 0.05 in Fig. 6(a) and 0.28 in Fig. 6(b).

For the external-only method in Fig. 6(a), it suggests that the estimated quality of correct triples [Dataset(A)-(a, b)] has a massive drop and the quality of incorrect triples [Dataset(A)-(c, d, e, f)] increases greatly to become higher than that of correct ones. This wrong evaluation result occurs since the incorrect triples are much more than the correct triples, thus greatly increasing the quality of incorrect triples that are consistent with each other and degrading the quality of correct but inconsistent ones.

For the proposed method in Fig. 6(a), it indicates that the quality of correct triples increases and that of incorrect ones decreases. The estimated quality of correct triples maintains higher than the quality of incorrect triples after iterations. It produces a much more reliable result than the external-only method, where the correct triple could be detected even in the existence of a large proportion of incorrect triple. The critical part is the knowledge verification from data. The correct triples usually have a stronger causal strength or correlation between their entities than incorrect ones. Through (10), (12), and (14), the weight of correct triples is increased to influence more on the quality evaluation. Therefore, correct triples could resist the influences from incorrect triples.

In Fig. 6(b), the proposed method also has a much better evaluation result than the external-only method. Moreover, it is worth noting that even though the quality of correct triples has a slight degradation, it still keeps higher than that of incorrect triples. The reason is that the causal strength of incorrect triples (0.28) here is not much lower than that of correct triples (0.38) as the example of Dataset (A). Even if under this condition, the verification of causality could still correct the KQ evaluation, indicating the effect of the verification mechanism of the proposed method. The above results demonstrate that the knowledge verification from data introduced in the proposed method could alleviate the interference of incorrect triples and produce reliable results even under the condition of large proportion of incorrect triples.

2) Performance of the Method Under Different Accuracy Conditions of Data Analysis: To show the influence of the accuracy of the data analysis on KQ evaluation, experiments are conducted with the external-only and proposed methods

TABLE I

PERFORMANCE OF THE PROPOSED METHOD AND THE EXTERNAL-ONLY METHOD UNDER DIFFERENT ACCURACY CONDITIONS OF CAUSALITY AND CORRELATION RESULTS

Accuracy condition /%		Dataset (A) /%					Dataset (B) /%				
Correlation	Causality	q_1	q_2	q_3	q_4	q_5	q_1	q_2	q_3	q_4	q_5
50	50	50.3+**12.3**	82.0+**5.4**	83.1+**4.6**	68.6+**10.4**	73.3+**6.7**	50.0+**9.6**	81.8+**2.0**	82.6+**0.9**	69.1+**8.0**	73.0+**1.9**
	60	49.5+**15.2**	81.7+**6.6**	84.2+**4.9**	68.4+**12.1**	72.4+**8.2**	50.0+**11.0**	81.7+**3.4**	82.7+**1.6**	68.9+**9.3**	73.3+**3.5**
	70	49.8+**16.6**	81.5+**7.9**	83.3+**6.4**	68.6+**13.1**	73.0+**10.4**	49.5+**12.9**	82.0+**4.5**	82.9+**3.2**	68.1+**11.2**	73.3+**5.4**
	80	49.8+**18.8**	81.8+**9.2**	83.0+**8.2**	68.3+**15.3**	72.8+**12.3**	50.5+**14.6**	81.8+**5.9**	82.9+**4.6**	68.8+**11.8**	73.3+**7.1**
	90	49.8+**20.7**	81.9+**10.4**	83.0+**9.4**	68.6+**16.0**	72.9+**13.6**	49.9+**17.1**	81.8+**7.2**	83.5+**5.6**	68.7+**13.2**	72.3+**9.4**
60	50	49.5+**12.8**	81.6+**5.3**	83.1+**4.7**	68.1+**10.4**	73.1+**6.5**	49.6+**9.9**	81.6+**2.3**	82.8+**0.6**	68.4+**8.4**	73.8+**3.2**
	60	50.2+**14.8**	82.2+**6.4**	83.3+**5.6**	69.1+**12.0**	73.6+**8.0**	50.2+**12.3**	81.5+**3.8**	82.6+**2.6**	69.2+**9.7**	73.0+**4.8**
	70	50.1+**17.0**	81.9+**8.0**	83.1+**6.8**	68.7+**13.5**	73.1+**10.6**	49.6+**14.4**	82.0+**4.6**	83.0+**3.1**	70.1+**10.4**	72.9+**6.4**
	80	49.6+**19.1**	81.9+**9.0**	83.0+**8.3**	68.4+**15.5**	72.2+**12.1**	49.8+**15.4**	81.5+**6.1**	83.2+**4.5**	68.6+**12.6**	72.5+**7.9**
	90	50.0+**21.1**	81.5+**10.7**	82.9+**9.5**	68.8+**16.5**	72.6+**14.4**	50.2+**16.9**	81.6+**7.7**	82.9+**5.9**	68.1+**13.4**	72.6+**9.7**
70	50	49.7+**13.5**	81.8+**6.0**	83.0+**5.1**	68.9+**10.9**	73.1+**6.8**	49.5+**10.8**	81.9+**3.0**	83.1+**1.5**	68.5+**8.5**	73.2+**3.6**
	60	49.7+**15.2**	81.9+**6.7**	83.5+**5.6**	68.4+**12.6**	73.6+**8.3**	50.0+**12.8**	81.7+**4.2**	83.2+**2.6**	68.9+**10.6**	72.7+**5.6**
	70	50.0+**16.9**	81.5+**8.0**	83.4+**6.8**	68.2+**13.6**	72.9+**10.3**	50.1+**15.3**	81.5+**5.9**	82.6+**4.2**	68.5+**12.4**	73.2+**7.5**
	80	49.9+**18.9**	82.2+**9.2**	82.2+**8.9**	68.4+**15.5**	73.9+**12.0**	49.7+**17.0**	81.6+**7.0**	82.9+**5.6**	69.2+**13.3**	72.3+**9.5**
	90	49.8+**21.3**	81.6+**10.8**	82.6+**10.1**	70.2+**15.6**	72.9+**14.0**	49.8+**18.7**	81.8+**8.3**	82.8+**6.8**	68.2+**15.2**	73.8+**10.1**
80	50	50.0+**13.2**	81.5+**5.9**	83.2+**5.0**	69.0+**11.0**	73.3+**7.3**	49.5+**11.7**	81.8+**3.7**	83.1+**1.9**	68.5+**9.8**	73.2+**3.9**
	60	50.1+**15.2**	81.5+**7.2**	83.3+**5.8**	68.6+**12.6**	72.8+**9.2**	49.8+**13.7**	81.7+**5.2**	83.1+**3.4**	69.2+**10.9**	74.3+**5.8**
	70	50.2+**17.0**	82.0+**8.2**	82.4+**7.1**	68.2+**14.0**	73.9+**10.1**	50.0+**15.6**	82.1+**6.1**	82.6+**5.1**	70.1+**11.8**	73.6+**8.1**
	80	50.1+**19.3**	81.5+**9.7**	82.9+**8.8**	68.4+**15.6**	72.9+**12.6**	50.0+**17.3**	81.5+**7.3**	83.3+**5.6**	68.5+**13.2**	72.9+**9.1**
	90	49.8+**21.3**	81.9+**10.7**	83.5+**9.7**	68.4+**16.7**	73.5+**14.0**	50.2+**19.1**	81.8+**8.5**	82.8+**7.4**	68.6+**14.6**	72.7+**11.0**
90	50	50.6+**13.1**	81.4+**6.2**	82.6+**5.1**	68.2+**11.0**	72.6+**7.3**	50.0+**12.0**	81.8+**4.4**	82.4+**3.4**	68.5+**10.4**	73.1+**5.3**
	60	49.7+**15.7**	81.6+**7.6**	83.0+**6.0**	67.9+**13.3**	73.3+**9.1**	50.1+**13.8**	81.8+**5.8**	82.6+**4.6**	68.6+**11.9**	72.6+**6.9**
	70	49.5+**17.6**	82.1+**8.4**	82.5+**7.6**	69.0+**13.5**	73.8+**10.9**	49.7+**16.1**	81.9+**6.7**	83.1+**5.3**	68.5+**13.0**	73.9+**8.5**
	80	50.4+**19.0**	82.1+**9.5**	83.0+**8.6**	68.2+**15.0**	73.9+**12.4**	50.1+**18.2**	81.9+**7.8**	83.0+**6.8**	68.8+**14.2**	73.4+**10.1**
	90	49.8+**21.4**	81.9+**10.9**	83.2+**9.9**	68.6+**16.7**	72.8+**14.3**	49.9+**20.4**	81.6+**9.4**	82.9+**8.0**	68.9+**15.7**	73.3+**12.5**

*To highlight the comparison of the two methods, the results are reported as the form of the accuracy of external-only method plus the improved accuracy of the proposed method. Take "81.3+5.5 (=86.8)" as an example, 81.3% is the accuracy of the external-only method, and 86.8% is the accuracy of the proposed method.

under different accuracy conditions of causality and correlation analysis. To simulate different accuracy conditions of correlation/causality in the multisource KQ evaluation scenario, partial correct (incorrect) results of the correlation and causality analysis method are random set to incorrect (correct) ones to fix their accuracy with a desired level. The experiments generate five group of multiple KGs on the basis of each dataset, and each set contains ten KGs. Five different sets of parameters are used to generate incorrect triples, which are $q_1 = \{0.57, 0.68, 0.4, 0.55, 0.34, 0.48, 0.68, 0.45, 0.63, 0.48\}$, $q_2 = \{0.21, 0.31, 0.25, 0.54, 0.61, 0.68, 0.23, 0.59, 0.66, 0.2\}$, $q_3 = \{0.54, 0.36, 0.42, 0.28, 0.26, 0.67, 0.44, 0.35, 0.26, 0.3\}$, $q_4 = \{0.48, 0.67, 0.45, 0.46, 0.51, 0.22, 0.61, 0.29, 0.44, 0.42\}$, and $q_5 = \{0.66, 0.33, 0.35, 0.56, 0.35, 0.36, 0.31, 0.5, 0.62, 0.33\}$. Each experiment is repeated ten times and the average performance of the proposed and external method is reported in Table I.

It suggests that the proposed method outperforms the external-only method in most conditions, even at the worst observed accuracy condition of causality and correlation. This result indicates that the verification could significantly enhance the effect of the KQ evaluation through integration with the consistency-based method, and the proposed method implements it in a proper way that could keep stable performance even with bad causality or correlation analysis.

Moreover, it also shows that the improved performance of the proposed method is more significant in the case of better causality and correlation analysis. This is because the correlation and causality extracted from data could provide more accurate verification, thus more accurately indicating the most probable correct knowledge. This capability to eliminate interference could guide the KQ evaluation in the right direction under the existence of much incorrect knowledge related to the same fact.

Note that the performance increases more significantly when the accuracy of causality analysis increases than that of the correlation analysis. This result represents that the causality is more critical to have more influence on the quality evaluation in the proposed method.

3) Influence of Data Noise on the Performance of the Proposed Method: To show how the method performs in the case of noisy data, the data with different degrees of noise are generated and the proposed method is applied together with the external-only method to evaluate KQ accordingly. Noisy data are generated according to a noise parameter λ, which represents the degree of noise. Specifically, the property of each sample has probability λ to be set to a different value to introduce noise. Eight groups of KGs (each contains ten KGs) are generated based on Dataset (A), and different groups simulate different multisource KGs scenarios, where experiments are conducted separately on these groups of KGs. For ablation experiments, the accuracy of using only correlation and causal strength to select high-quality knowledge is also presented. The result of each group of experiment is shown in Fig. 7. Moreover, the corresponding causal strength between all entities is presented as the form of matrix in Fig. 8

It suggests that as the noise increases, the accuracy of the method using correlation and causality tends to degrade. Even if the accuracy of the proposed method also decreases as the noise increases, it keeps in a much more stable level and maintains a higher accuracy than the external-only method. The good resistance to data noise is due to the proper way

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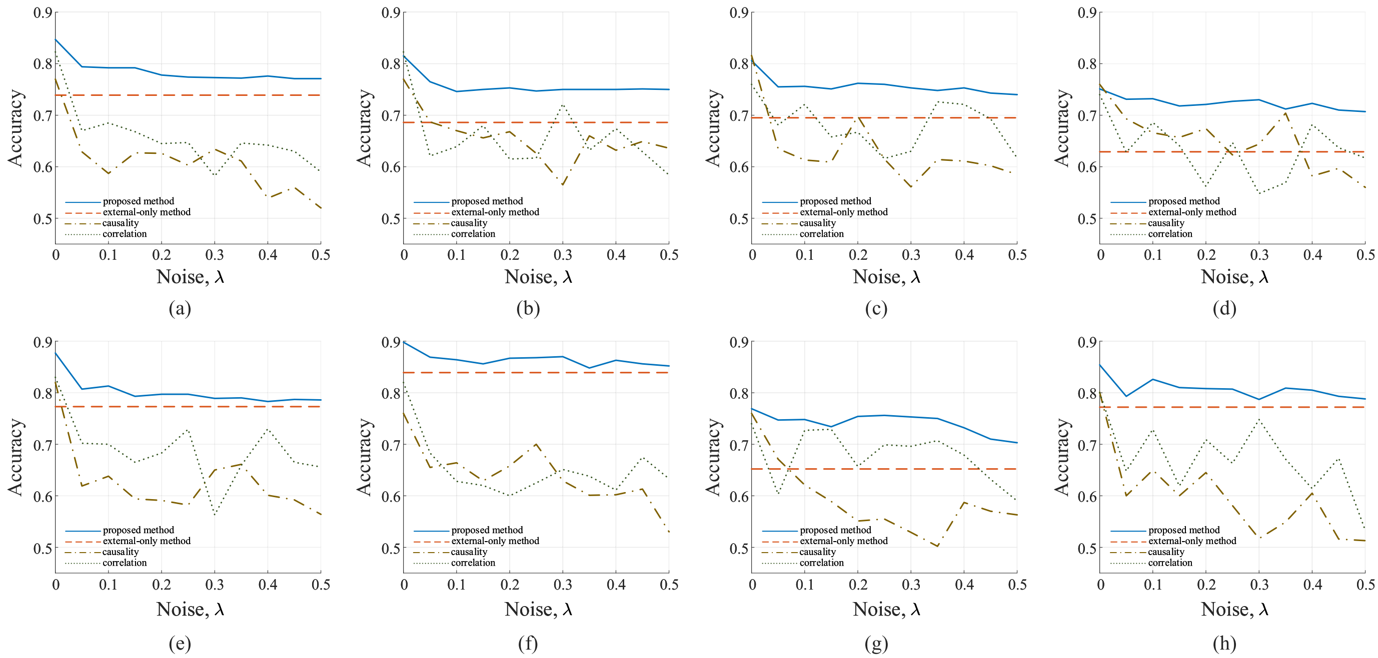

Fig. 7. Accuracy of different methods under different noise conditions. *Each experiment shares the same generated KGs, and it is why the accuracy of the external-only method does not change. (a) ext_acc = 0.739. (b) ext_acc = 0.686. (c) ext_acc = 0.695. (d) ext_acc = 0.622. (e) ext_acc = 0.772. (f) ext_acc = 0.839. (g) ext_acc = 0.652. (h) ext_acc = 0.787.

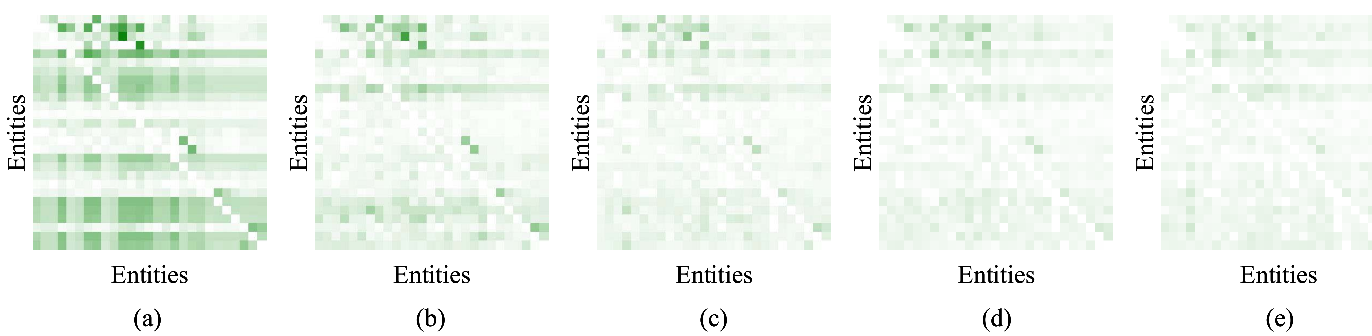

Fig. 8. Matrix of the causal strength between all entities under different conditions of noise. (a) Noise, $\lambda = 0.1$. (b) Noise, $\lambda = 0.2$. (c) Noise, $\lambda = 0.3$. (d) Noise, $\lambda = 0.4$. (e) Noise, $\lambda = 0.5$.

to integrate causality and correlation information into the consistency-based iterations. When the noise of data comes to a certain extent, the method could reach a balance of referring to consistency comparison or verification of data. The interference of KQ evaluation caused by data noise could be alleviated through the correction of consistency information, and the part of correct data analysis still works for the improvement of KQ evaluation.

More detailed analysis of how the multisource consistency resists noise could be seen from Fig. 8, where darker color represents a larger value. It shows that as the noise increases, the causal strength tends to be degraded to 0 but not to be messed up in the order of values. For the triples whose causal strength between entities is greatly degraded, the data verification falls ineffective on the quality evaluation of them but will not mislead the evaluation based on the multisource consistency. The proposed method on these triples could be regarded as degenerate to the external-only method. For the triples whose causal strength between entities is not degraded so much by the noise, their causal strength is still greater than that of incorrect triples, and therefore, the verification from data on these triples still plays a role to improve the evaluation results. Through the above process, the proposed method could still work even under greatly noisy data.

4) Impact of the Number of KGs: To show the influence of the number of KGs on the proposed method, experiments are performed on different numbers of KGs based on Dataset (B). Since the number of KGs mainly affects the effect of the consistency-based part of the proposed method, the accuracy of other consistency-based methods, such as external-only method and expectation-maximization (EM) algorithm [68], is also reported in Fig. 9 for better analysis. There are four parameters that generate incorrect triples utilized for this experiment, which are $s_1 = \{0.67, 0.4, 0.7, 0.55, 0.24\}$, $s_2 = \{0.51, 0.5, 0.26, 0.46, 0.26\}$, $s_3 = \{0.4, 0.67, 0.52, 0.31, 0.44\}$, and $s_4 = \{0.59, 0.3, 0.58, 0.58, 0.29\}$.

TABLE II

PERFORMANCE OF THE PROPOSED METHOD AND OTHER METHODS

Datasets	Dataset (A) /%				Dataset (B) /%				Dataset (C) /%				Dataset (D) /%				Friedman rank
Parameters	p_1	p_2	p_3	p_4	p_1	p_2	p_3	p_4	p_1	p_2	p_3	p_4	p_1	p_2	p_3	p_4	
Weighted Majority Vote [69]	85.4	76.6	79.0	71.2	85.0	77.8	79.5	75.1	85.4	79.4	79.8	71.9	86.6	80.7	77.8	69.0	3.75
Majority Vote [70]	82.1	66.9	59.7	59.0	81.6	67.1	56.7	56.1	82.3	68.9	60.3	61.5	81.6	70.5	58.0	57.1	5.00
EM algorithm [68]	97.3	90.1	87.2	84.5	96.9	90.9	86.2	82.3	96.4	91.2	87.6	85.3	96.5	88.3	87.5	83.0	2.00
External-only method	88.3	81.4	73.6	73.9	89.3	79.80	75.2	73.4	90.4	85.8	78.0	80.0	91.8	85.8	79.3	79.6	3.25
Proposed method	**99.3**	**93.2**	**91.6**	**88.8**	**99.1**	**94.2**	**93.0**	**89.8**	**99.4**	**97.8**	**97.1**	**97.4**	**99.9**	**99.2**	**99.3**	**99.1**	1.00

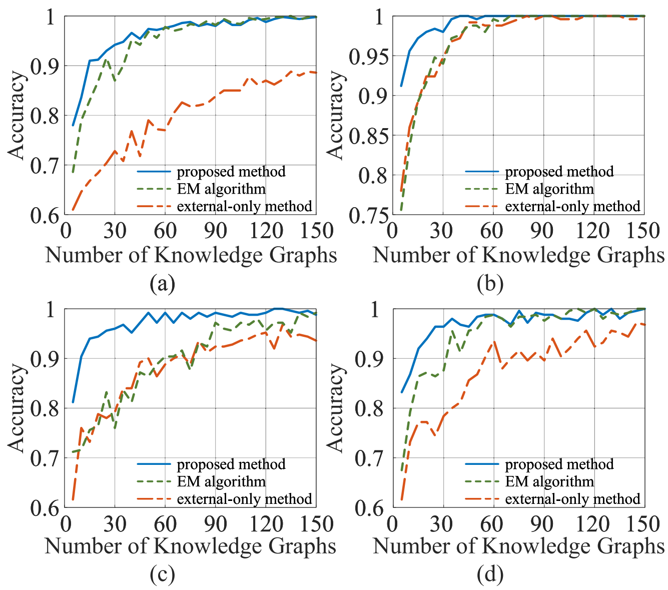

Fig. 9. Accuracy of the proposed, external-only method, and EM algorithm in the case of different numbers of KGs. (a) Parameter: s_1. (b) Parameter: s_2. (c) Parameter: s_3. (d) Parameter: s_4.

It suggests that as the number of external knowledge increases, the performance of all the three methods improves, and the proposed method outperforms the other ones (methods based on multisource consistency). Under the four parameters, the performance of the external-only method is lower than that of the EM algorithm for s_1 and s_4, and they are close for s_2 and s_3. For all cases, the proposed method is the best in performance, which demonstrates that the verification from data is significant for knowledge validation as supplementary information for the multisource consistency.

Specifically, when the number of external triples is small, the improved performance of the proposed method is especially significant. This is because the small amount of external knowledge leads to a greater probability that the knowledge describing a fact has a large proportion of incorrect knowledge. In this case, the verification of the proposed method could produce a significant correction of knowledge weights, thus reducing the error evaluation in the condition of much incorrect knowledge to a great extent.

In real-world cases of knowledge verification, the number of available knowledge sources tends to be small due to the expensive cost of collecting multisource knowledge. For example, in the crowdsourcing scenario, workers that construct knowledge for certain facts could be regarded as knowledge sources. Knowledge could be assigned to several or tens of workers to process, that is, there may be at most dozens of available knowledge sources. In this case, the proposed method and even the external-only method (the proposed method without verification from data) greatly outperform the traditional EM method. Moreover, the verification from data is also especially effective in improving the accuracy, by most 10% within 50 KGs. It is worth noting that the performance of the proposed method increases more stably than the other methods, indicating that the verification from data could defend against the inference caused by incorrect knowledge.

In general, the proposed method performs better with more KGs, which is due to the improving performance of the consistency-based part with more KGs. Besides, it demonstrates that knowledge verification from data has an especially significant effect to assist in proper KQ evaluation in the case of small number of KGs.

V. DISCUSSION

A. Comparison With Other Methods

To show the effect of introducing knowledge verification from data to KQ evaluation, the proposed method is compared with other three multiconsistency-based methods, i.e., majority vote, weighted majority vote, and EM algorithm and the variant of the proposed method and external-only method. Four groups of KGs (each group contains five KGs) are generated with different probabilities p. Specifically, the probability p of each group of KGs is set as $p_1 = \{0.01, 0.1, 0.3, 0.5, 0.7\}$, $p_2 = \{0.1, 0.2, 0.3, 0.5, 0.7\}$, $p_3 = \{0.1, 0.3, 0.4, 0.5, 0.7\}$, and $p_4 = \{0.2, 0.2, 0.4, 0.5, 0.7\}$.

The performance of the proposed method and the three comparison methods is shown in Table II. For statistical analysis of the performance, the Friedman test with the Wilcoxon signed-rank *post hoc* tests [71] is utilized. They are conducted based on the rank of each method under every parameter setting p of each dataset (16 in total). The mean rank of each method is also presented in Table II. According to the Friedman test, a significant difference statistically exists in the effectiveness of the five methods, $\chi^2(15) = 61.6$, $p < 0.001$. The Wilcoxon signed-rank *post hoc* tests subsequently indicate that the proposed method significantly outperforms the other four methods, i.e., weighted majority vote ($Z = 4.8$, $p < 0.001$), majority vote ($Z = 4.8$, $p < 0.001$), EM algorithm ($Z = 3.6$, $p < 0.001$), and external-only method ($Z = 4.6$, $p < 0.001$).

The statistical analysis suggests that the proposed method outperforms other three consistency-based methods in all conditions of parameters, which demonstrates that the introduction of knowledge verification from data is effective to assist in the KQ evaluation.

Suppose that there are N KGs, M triples in each KG and L iterations for all iterative methods. The time complexity of the proposed method is $O(LMN^2)$. It is definitely slower than the baselines, EM algorithm with $O(LMN)$ and MV algorithm with $O(N)$. However, as mentioned in Section IV-C4, the available KGs are usually limited in a small number in real-world scenarios, i.e., N is usually a small number. In this context, the consumed time of the proposed method is acceptable. Moreover, in comparison to the EM algorithm that is in serial logic, the proposed method could be implemented in a parallel way since the calculation of the next iteration only depends on the outcomes of the previous iteration. Therefore, the consumed time of the proposed method could be further optimized to promise its feasibility in practice.

B. Limitation and Future Work

Data contain a plenty of objective records about real-word facts, which has great potential to assist in KG quality management but is seldom explored. As an exploration, this article employs a causal analysis of data properties to mine the data-driven knowledge and assist the KQ evaluation in the multisource scenarios. However, the analysis of causality can only verify whether there might be a relationship between entities but cannot detect whether the specific relationship is correct. Therefore, it is significant for the promotion and utility of data in KG quality management to explore a more fine-grained framework of verification on quality evaluation. For example, it could be considered to classify relationships in some ways and analyze the effect that the statistics validates different classes of relationships.

This article validates knowledge by mining knowledge-like information from data, which is actually an attempt to discover potential correlations of data and knowledge. In the context of surging research on explainable artificial intelligence (XAI), this idea could become an enlightening way to reveal the human-understandable mechanism of data learning models. For example, could the causal or correlation analysis be integrated in the model design or model interpretation process to construct an explainable baseline or give a reasonable illustration on the prediction results? Hopefully, this article could facilitate further exploration of data and knowledge in the XAI.

C. Feasibility and Extensibility

The proposed method incorporates the verification from data with the multisource consistency to improve the knowledge validation process. Experiments show that the verification from data is effective to giving a more accurate evaluation of KQ. Especially under the usual condition of small number of KGs available for multisource consistency comparison, the verification from data significantly addresses the confusing evaluation situations. Since the causal and correlation analysis cooperates to verify knowledge validation, the verification form data could benefit the evaluation of knowledge with not only causal relationships but also any form of correlation relationship. Moreover, even if the correlation analysis results between the entities are wrong, the multi-source consistency could also work well in ordinary cases to evaluate the KQ, as shown in the ablation experiments (Sections IV-C2 and IV-C3).

As for the extensibility, the proposed method processes data with discrete values for causal and correlation analysis. It could easily be extended to deal with continuous data by using existing statistical approaches to analyze correlation degree and causal strength from the continuous data. For correlation degree of continuous data, discretization-based tests [72], regression-based tests [73], and kernel-based tests [74], could be used. For causal strength, preliminary causal discovery methods [75], [76] for continuous data and subsequent causal inference methods for continuous data [77], [78], [79] could be used to calculate the causal strengths proposed in this article.

VI. Conclusion

This article explores a way to mine the hidden knowledge in the numerical data to verify the triples in KGs, and a method is proposed to introduce causality and correlation of data to the consistency-based method to evaluate the quality of triples. Experiments are conducted in multisource scenario to analyze the effect of data verification on the quality evaluation. The results show that knowledge verification from data is effective to alleviate the interference of incorrect knowledge and improves the accuracy of the consistency-based method. It explores a new way of interaction between knowledge and data and proves the feasibility of data-driven information assisting the KG quality management. Based on this exploration, how to make better utilization of existing large-scaled, low-cost data to assist high-quality KG construction and knowledge applications may have important research value in the future.

References

[1] X. Zhao, H. Chen, Z. Xing, and C. Miao, "Brain-inspired search engine assistant based on knowledge graph," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Oct. 5, 2021, doi: 10.1109/TNNLS.2021.3113026.

[2] T. Ban, X. Wang, L. Chen, X. Wu, Q. Chen, and H. Chen, "Quality evaluation of triples in knowledge graph by incorporating internal with external consistency," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jul. 4, 2022, doi: 10.1109/TNNLS.2022.3186033.

[3] G. Leu and H. Abbass, "A multi-disciplinary review of knowledge acquisition methods: From human to autonomous eliciting agents," *Knowl.-Based Syst.*, vol. 105, no. 1, pp. 1–22, Aug. 2016.

[4] X. Wang, A. Chused, N. Elhadad, C. Friedman, and M. Markatou, "Automated knowledge acquisition from clinical narrative reports," in *Proc. Amer. Med. Informat. Assoc. Annu. Symp.*, 2008, pp. 783–787.

[5] A. H. Mohammad and N. A. M. A. Saiyd, "A framework for expert knowledge acquisition," *Int. J. Comput. Sci. Netw. Secur.*, vol. 10, no. 11, p. 145, 2010.

[6] W. Zhang, H. Xu, and W. Wan, "Weakness finder: Find product weakness from Chinese reviews by using aspects based sentiment analysis," *Expert Syst. Appl.*, vol. 39, no. 11, pp. 10283–10291, 2012.

[7] S.-H. Liao, "Expert system methodologies and applications—A decade review from 1995 to 2004," *Expert Syst. Appl.*, vol. 28, no. 1, pp. 93–103, Jan. 2005.

[8] X. Chu *et al.*, "KATARA: A data cleaning system powered by knowledge bases and crowdsourcing," in *Proc. ACM SIGMOD Int. Conf. Manag. Data*, May 2015, pp. 1247–1261.

[9] M. Trokhymovych and D. Saez-Trumper, "WikiCheck: An end-to-end open source automatic fact-checking API based on Wikipedia," in *Proc. 30th ACM Int. Conf. Inf. Knowl. Manag.*, Oct. 2021, pp. 4155–4164.

[10] A. Velu, "The spread of big data science throughout the globe," *Int. J. Sustain. Develop. Comput. Sci.*, vol. 1, no. 1, pp. 11–20, 2019.

[11] A. J. Medford, M. R. Kunz, S. M. Ewing, T. Borders, and R. Fushimi, "Extracting knowledge from data through catalysis informatics," *Amer. Chem. Soc. Catal.*, vol. 8, no. 8, pp. 7403–7429, Aug. 2018.

[12] H. Xiong, G. Pandey, M. Steinbach, and V. Kumar, "Enhancing data analysis with noise removal," *IEEE Trans. Knowl. Data Eng.*, vol. 18, no. 3, pp. 304–319, Mar. 2006.

[13] J. Pearl, *Causality*. Cambridge, U.K.: Cambridge Univ. Press, 2009.

[14] R. Taylor, "Interpretation of the correlation coefficient: A basic review," *J. Diagnostic Med. Sonogr.*, vol. 6, no. 1, pp. 35–39, 1990.

[15] M. Studeny, *Probabilistic Conditional Independence Structures*. London, U.K.: Springer, 2006.

[16] X. Wu, B. Jiang, K. Yu, C. Miao, and H. Chen, "Accurate Markov boundary discovery for causal feature selection," *IEEE Trans. Cybern.*, vol. 50, no. 12, pp. 4983–4996, Dec. 2020.

[17] X. Wu, B. Jiang, K. Yu, and H. Chen, "Separation and recovery Markov boundary discovery and its application in EEG-based emotion recognition," *Inf. Sci.*, vol. 571, pp. 262–278, Sep. 2021.

[18] M. Scanagatta, A. Salmerón, and F. Stella, "A survey on Bayesian network structure learning from data," *Prog. Artif. Intell.*, vol. 8, no. 4, pp. 425–439, 2019.

[19] W. Wang, H. Wang, B. Yang, L. Liu, P. Liu, and G. Zeng, "A Bayesian network-based knowledge engineering framework for IT service management," *IEEE Trans. Services Comput.*, vol. 6, no. 1, pp. 76–88, 1st Quart., 2013.

[20] A. C. Constantinou, Z. Guo, and N. K. Kitson, "The impact of prior knowledge on causal structure learning," 2021, *arXiv:2102.00473*.

[21] X. Wang *et al.*, "Knowledge graph quality control: A survey," *Fundam. Res.*, vol. 1, no. 5, pp. 607–626, Sep. 2021.

[22] M. H. Gad-Elrab, D. Stepanova, J. Urbani, and G. Weikum, "Tracy: Tracing facts over knowledge graphs and text," in *Proc. World Wide Web Conf.*, 2019, pp. 3516–3520.

[23] J. Lajus and F. M. Suchanek, "Are all people married? Determining obligatory attributes in knowledge bases," in *Proc. World Wide Web Conf. World Wide Web*, 2018, pp. 1115–1124.

[24] F. Li, X. L. Dong, A. Langen, and Y. Li, "Knowledge verification for long-tail verticals," *Proc. VLDB Endowment*, vol. 10, no. 11, pp. 1370–1381, Aug. 2017.

[25] B. Xue and L. Zou, "Knowledge graph quality management: A comprehensive survey," *IEEE Trans. Knowl. Data Eng.*, early access, Feb. 10, 2022, doi: 10.1109/TKDE.2022.3150080.

[26] P. Ojha and P. Talukdar, "KGEval: Accuracy estimation of automatically constructed knowledge graphs," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2017, pp. 1741–1750.

[27] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," 2013, *arXiv:1301.3781*.

[28] L. Galárraga, S. Razniewski, A. Amarilli, and F. M. Suchanek, "Predicting completeness in knowledge bases," in *Proc. 10th ACM Int. Conf. Web Search Data Mining*, Feb. 2017, pp. 375–383.

[29] A. Liew, "Understanding data, information, knowledge and their inter-relationships," *J. Knowl. Manag. Pract.*, vol. 8, no. 2, pp. 1–16, 2007.

[30] A. P. Dawid, "Conditional independence in statistical theory," *J. Roy. Stat. Soc. B, Methodol.*, vol. 41, no. 1, pp. 1–15, Sep. 1979.

[31] J. H. McDonald, *Handbook of Biological Statistics*, vol. 2. Baltimore, MD, USA: Sparky House Publishing, 2009.

[32] J. M. Pena, "Learning Gaussian graphical models of gene networks with false discovery rate control," in *Proc. Eur. Conf. Evol. Comput., Mach. Learn. Data Mining Bioinf.*, 2008, pp. 165–176.

[33] K. Zhang, B. Schölkopf, P. Spirtes, and C. Glymour, "Learning causality and causality-related learning: Some recent progress," *Nat. Sci. Rev.*, vol. 5, no. 1, pp. 26–29, 2018.

[34] L. Yao, Z. Chu, S. Li, Y. Li, J. Gao, and A. Zhang, "A survey on causal inference," *ACM Trans. Knowl. Discovery Data*, vol. 15, no. 5, pp. 1–46, 2021.

[35] X. Wu, B. Jiang, K. Yu, H. Chen, and C. Miao, "Multi-label causal feature selection," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 4, pp. 6430–6437.

[36] D. Yang, H. Chen, Y. Song, and Z. Gong, "Granger causality for multivariate time series classification," in *Proc. IEEE Int. Conf. Big Knowl. (ICBK)*, Aug. 2017, pp. 103–110.

[37] X. Wu, B. Jiang, Y. Zhong, and H. Chen, "Multi-label causal variable discovery: Learning common causal variables and label-specific causal variables," 2020, *arXiv:2011.04176*.

[38] C. Glymour, K. Zhang, and P. Spirtes, "Review of causal discovery methods based on graphical models," *Frontiers Genet.*, vol. 10, p. 524, Jun. 2019.

[39] J. Abellán *et al.*, "Some variations on the PC algorithm," in *Probabilistic Graphical Models*. Prague, Czech Republic: Zeithamlová Milena, Ing.-Agentura Action M, 2006, pp. 1–8.

[40] P. Spirtes, C. N. Glymour, R. Scheines, and D. Heckerman, *Causation, Prediction, and Search*. Cambridge, MA, USA: MIT Press, 2000.

[41] D. M. Chickering, "Optimal structure identification with greedy search," *J. Mach. Learn. Res.*, vol. 3, pp. 507–554, Nov. 2002.

[42] M. J. Vowels, N. C. Camgoz, and R. Bowden, "D'ya like DAGs? A survey on structure learning and causal discovery," 2021, *arXiv:2103.02582*.

[43] S. Shimizu, P. O. Hoyer, A. Hyvärinen, A. Kerminen, and M. Jordan, "A linear non-Gaussian acyclic model for causal discovery," *J. Mach. Learn. Res.*, vol. 7, no. 10, pp. 2003–2030, 2006.

[44] P. O. Hoyer, S. Shimizu, A. J. Kerminen, and M. Palviainen, "Estimation of causal effects using linear non-Gaussian causal models with hidden variables," *Int. J. Approx. Reasoning*, vol. 49, no. 2, pp. 362–378, 2008.

[45] J. Choi, R. Chapkin, and Y. Ni, "Bayesian causal structural learning with zero-inflated Poisson Bayesian networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 5887–5897.

[46] N. Friedman and D. Koller, "Being Bayesian about network structure. A Bayesian approach to structure discovery in Bayesian networks," *Mach. Learn.*, vol. 50, nos. 1–2, pp. 95–125, 2003.

[47] I. Tsamardinos, L. E. Brown, and C. F. Aliferis, "The max-min hill-climbing Bayesian network structure learning algorithm," *Mach. Learn.*, vol. 65, no. 1, pp. 31–78, 2006.

[48] J. Pearl, M. Glymour, and N. P. Jewell, *Causal Inference in Statistics: A Primer*. Hoboken, NJ, USA: Wiley, 2016.

[49] E. A. Stuart, "Matching methods for causal inference: A review and a look forward," *Stat. Sci.*, vol. 25, no. 1, pp. 1–21, Feb. 2010.

[50] S. Athey and G. Imbens, "Recursive partitioning for heterogeneous causal effects," *Proc. Nat. Acad. Sci. USA*, vol. 113, no. 27, pp. 7353–7360, Jul. 2016.

[51] S. Athey, M. Bayati, G. Imbens, and Z. Qu, "Ensemble methods for causal effects in panel data settings," *Amer. Econ. Assoc. Papers Proc.*, vol. 109, pp. 65–70, May 2019.

[52] M.-C. Yuen, I. King, and K.-S. Leung, "A survey of crowdsourcing systems," in *Proc. IEEE 3rd Int. Conf. Privacy, Secur., Risk Trust IEEE 3rd Int. Conf. Social Comput.*, Oct. 2011, pp. 766–773.

[53] J. Pasternack and D. Roth, "Generalized fact-finding," in *Proc. 20th Int. Conf. Companion World Wide Web*, 2011, pp. 99–100.

[54] D. Tao, J. Cheng, Z. Yu, K. Yue, and L. Wang, "Domain-weighted majority voting for crowdsourcing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 1, pp. 163–174, Jan. 2018.

[55] D. Gerber *et al.*, "DeFacto: Temporal and multilingual deep fact validation," *J. Web Semantics*, vol. 35, pp. 85–101, Dec. 2015.

[56] X. Yin, J. Han, and P. S. Yu, "Truth discovery with multiple conflicting information providers on the web," *IEEE Trans. Knowl. Data Eng.*, vol. 20, no. 6, pp. 796–808, Jun. 2008.

[57] J. Pasternack and D. Roth, "Making better informed trust decisions with generalized fact-finding," in *Proc. 22nd Int. Joint Conf. Artif. Intell.*, 2011, pp. 2324–2329.

[58] J. Pasternack and D. Roth, "Latent credibility analysis," in *Proc. 22nd Int. Conf. World Wide Web*, 2013, pp. 1009–1020.

[59] H. Chen, G. Cao, J. Chen, and J. Ding, "A practical framework for evaluating the quality of knowledge graph," in *Proc. China Conf. Knowl. Graph Semantic Comput.*, 2019, pp. 111–122.

[60] P. Dalgaard, "Power and the computation of sample size," in *Introductory Statistics With R*. New York, NY, USA: Springer, 2008, pp. 155–162.

[61] E. B. Wilson and M. M. Hilferty, "The distribution of chi-square," *Proc. Nat. Acad. Sci. USA*, vol. 17, no. 12, p. 684, 1931.

[62] Q. Liu, J. Peng, and A. T. Ihler, "Variational inference for crowdsourcing," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012, pp. 1–9.

[63] J. Binder, D. Koller, S. Russell, and K. Kanazawa, "Adaptive probabilistic networks with hidden variables," *Mach. Learn.*, vol. 29, no. 2, pp. 213–244, 1997.

[64] W. Wang *et al.*, "Prior-knowledge-driven local causal structure learning and its application on causal discovery between type 2 diabetes and bone mineral density," *IEEE Access*, vol. 8, pp. 108798–108810, 2020.

[65] I. A. Beinlich, H. J. Suermondt, R. M. Chavez, and G. F. Cooper, "The alarm monitoring system: A case study with two probabilistic inference techniques for belief networks," in *Proc. 2nd Eur. Conf. Artif. Intell. Med.*, 1989, pp. 247–256.

[66] B. Abramson, J. Brown, W. Edwards, A. Murphy, and R. L. Winkler, "Hailfinder: A Bayesian system for forecasting severe weather," *Int. J. Forecasting*, vol. 12, no. 1, pp. 57–71, Mar. 1996.

[67] F. Rodrigues, F. Pereira, and B. Ribeiro, "Sequence labeling with multiple annotators," *Mach. Learn.*, vol. 95, no. 2, pp. 165–181, May 2014.

[68] D. R. Karger, S. Oh, and D. Shah, "Iterative learning for reliable crowdsourcing systems," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 24, 2011, pp. 1–9.

[69] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, and O. Yakhnenko, "Translating embeddings for modeling multi-relational data," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 26, 2013, pp. 1–9.

[70] W. Tang and M. Lease, "Semi-supervised consensus labeling for crowdsourcing," in *Proc. Special Interest Group Inf. Retr. Workshop Crowdsourcing Inf. Retr.*, 2011, pp. 1–6.

[71] W. Min, E. Y. Ha, J. Rowe, B. Mott, and J. Lester, "Deep learning-based goal recognition in open-ended digital games," in *Proc. 10th AAAI Conf. Artif. Intell. Interact. Digit. Entertainment*, 2014, pp. 37–43.

[72] T.-M. Huang, "Testing conditional independence using maximal nonlinear conditional correlation," *Ann. Statist.*, vol. 38, no. 4, pp. 2047–2091, Aug. 2010.

[73] R. D. Shah and J. Peters, "The hardness of conditional independence testing and the generalised covariance measure," *Ann. Statist.*, vol. 48, no. 3, pp. 1514–1538, Jun. 2020.

[74] E. V. Strobl, K. Zhang, and S. Visweswaran, "Approximate kernel-based conditional independence tests for fast non-parametric causal discovery," *J. Causal Inference*, vol. 7, no. 1, pp. 1–24, Apr. 2019.

[75] J. Mooij, D. Janzing, J. Peters, and B. Schölkopf, "Regression by dependence minimization and its application to causal inference in additive noise models," in *Proc. 26th Annu. Int. Conf. Mach. Learn.*, 2009, pp. 745–752.

[76] E. Sokolova, P. Groot, T. Claassen, and T. Heskes, "Causal discovery from databases with discrete and continuous variables," in *Proc. Eur. Workshop Probabilistic Graph. Models*, 2014, pp. 442–457.

[77] F. P. Hartwig, "A generalized definition of the average causal effect for both binary and continuous treatments," 2021, *arXiv:2112.04580*.

[78] Y. Kitazawa, "Estimating the average causal effect of intervention in continuous variables using machine learning," 2022, *arXiv:2203.03916*.

[79] D. Galagate, "Causal inference with a continuous treatment and outcome: Alternative estimators for parametric dose-response functions with applications," Ph.D. dissertation, Dept. Math., Univ. Maryland, College Park, MD, USA, 2016.

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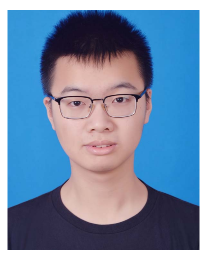

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